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Relational Algebra

GATE CSE Database Management System · Previous Year Questions with Answer Key

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Marks 1 16 questions

Q1. GATE CSE 2024 Set 1

Open solution

Consider the following two relations, **R(A, B)** and **S(A, C)**:

R	
A	**B**
10	20
20	30
30	40
30	50
50	95

S	
A	**C**
10	90
30	45
40	80

The total number of tuples obtained by evaluating the following expression

$$\sigma_{B<C}(R \bowtie_{R.A=S.A} S)$$

is _____

Q2. GATE CSE 2022

Open solution

Consider the following three relations in a relational database.

Employee ($\underline{eld}$, Name), Brand ($\underline{bld}$, bName), Own ($\underline{eld}$, $\underline{bld}$)

Which of the following relational algebra expressions return the set of elds who own all the brands?

(A) $\pi_{\text{eld}}(\pi_{\text{eld, bld}}(\text{Own}) / \pi_{\text{bld}}(\text{Brand}))$

(B) $\pi_{\text{eld}}(\text{Own}) - \pi_{\text{eld}}((\pi_{\text{eld}}(\text{Own}) \times \pi_{\text{bld}}(\text{Brand})) - \pi_{\text{eld, bld}}(\text{Own}))$

(C) $\pi_{\text{eld}}(\pi_{\text{eld, bld}}(\text{Own}) / \pi_{\text{bld}}(\text{Own}))$

(D) $\pi_{\text{eld}}((\pi_{\text{eld}}(\text{Own}) \times \pi_{\text{bld}}(\text{Own}) / \pi_{\text{bld}}(\text{Brand}))$

Q3. GATE CSE 2021 Set 1

Open solution

A relation r(A, B) in a relational database has 1200 tuples. The attribute A has integer values ranging from 6 to 20, and the attribute B has integer values ranging from 1 to 20. Assume that the attributes A and B are independently distributed. The estimated number of tuples in the output of $\sigma_{(A>10) \vee (B=18)}(r)$ is _____

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Q4. GATE CSE 2014 Set 3

What is the optimized version of the relation algebra expression $\pi_{A1}(\pi_{A2}(\sigma_{F1}(\sigma_{F2}(r))))$, where $A1, A2$ are sets of attributes in r with $A1 \subset A2$ and $F1, F2$ are Boolean expressions based on the attributes in r?

Open solution

(A) $\pi_{A1}(\sigma_{(F1 \wedge F2)}(r))$
(B) $\pi_{A1}(\sigma_{(F1 \vee F2)}(r))$
(C) $\pi_{A2}(\sigma_{(F1 \wedge F2)}(r))$
(D) $\pi_{A2}(\sigma_{(F1 \vee F2)}(r))$

Q5. GATE CSE 2008

Which of the following tuple relational calculus expression(s) is/are equivalent to $\forall t \in r\,(P\,(t))$?

I. $\neg \exists t \in r\,(P\,(t))$
II. $\exists t \notin r\,(P\,(t))$
III. $\neg \exists t \in r\,(\neg P\,(t))$
IV. $\exists t \notin r\,(\neg P\,(t))$

Open solution

(A) I only
(B) II only
(C) III only
(D) III and IV only

Q6. GATE CSE 2006

Consider the relations r_1(P, Q, R) and r_2(R, S, T) with primary keys P and R respectively. The relation r_1 contains 2000 tuples and r_2 contains 2500 tuples. The maximum size of the join $r_1 \bowtie r_2$ is

Open solution

(A) 2000
(B) 2500
(C) 4500
(D) 5000

Q7. GATE CSE 2006

Which of the following relational query languages have the same expressive power?
I) Relational algebra
II) Tuple relational calculus restricted to safe expressions
III) Domain relational calculus restricted to safe expressions

Open solution

(A) II and III only
(B) I and II only
(C) I and III only
(D) I, II and III

Q8. GATE CSE 2005

Let r be a relation instance with schema R = (A, B, C, D).

We define $r_1 = \pi_{A,B,C}\,(r)$ and $r_2 = \pi_{A,D}\,(r)$. Let $s = r_1 * r_2$ where * denotes natural join. Given that the decomposition of r into r_1 and r_2 is lossy, which one of the following is TRUE?

Open solution

(A) $s \subset r$
(B) $r \cup s = r$
(C) $r \subset s$
(D) $r * s = s$

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Q9. GATE CSE 2004

Open solution

Let R_1 (A, B, C) and R_2 (D, E) be two relation schema, where the primary keys are shown underlined, and let C be a foreign key in R_1 referring to R_2. Suppose there is no violation of the above referential integrity constraint in the corresponding relation instances r_1 and r_2. Which one of the following relational algebra expressions would necessarily produce an empty relation?

(A) $\prod_D (r_2) - \prod_C (r_1)$

(B) $\prod_C (r_1) - \prod_D (r_2)$

(C) $\prod_D (r_1 \bowtie_{C \neq D} r_2)$

(D) $\prod_C (r_1 \bowtie_{C = D} r_2)$

Q10. GATE CSE 2003

Open solution

Consider the following SQL query:

```
Select distinct a1, a2, ..., an

From r1, r2, ..., rm

Where P;
```

For an arbitrary predicate P, this query is equivalent to which of the following relational algebra expressions?

(A) $\prod_{a_1, a_2, \ldots, a_n} \sigma_p (r_1 \times r_2 \times \ldots \times r_m)$

(B) $\prod_{a_1, a_2, \ldots, a_n} \sigma_p (r_1 \bowtie r_2 \bowtie \ldots \bowtie r_m)$

(C) $\prod_{a_1, a_2, \ldots, a_n} \sigma_p (r_1 \cup r_2 \cup \ldots \cup r_m)$

(D) $\prod_{a_1, a_2, \ldots, a_n} \sigma_p (r_1 \cap r_2 \cap \ldots \cap r_m)$

Q11. GATE CSE 2002

Open solution

With regard to the expressive power of the formal relational query languages, which of the following statements is true?

(A) Relational algebra is more powerful than relational calculus.

(B) Relational algebra has the same power as relational calculus.

(C) Relational algebra has the same power as safe relational calculus.

(D) None of the above

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Q12. GATE CSE 2000

Open solution

Given the relations

employee (name, salary, deptno), and
department (deptno, deptname, address)

Which of the following queries cannot be expressed using the basic relational algebra operations $(\sigma, \pi, \times, \bowtie, \cup, \cap, -)$?

(A) Department address of every employee

(B) Employees whose name is the same as their department name

(C) The sum of all employee's salaries

(D) All employees of a given department

Q13. GATE CSE 1999

Open solution

Consider the join of a relation R with a relation S. If R has m tuples and S has n tuples then the maximum and minimum sizes of the join respectively are

(A) m + n and 0

(B) mn and 0

(C) m + n and |m − n|

(D) mn and m + n

Q14. GATE CSE 1999

Open solution

The relational algebra expression equivalent to the following tuple calculus expression:

$\{t | t \in r \wedge (t[A] = 10 \wedge t[B] = 20)\}$ is:

(A) $\sigma_{(A=10 \vee B=20)}(r)$

(B) $\sigma_{(A=10)}(r) \cup \sigma_{(B=20)}(r)$

(C) $\sigma_{(A=10)}(r) \cap \sigma_{(B=20)}(r)$

(D) $\sigma_{(A=10)}(r) - \sigma_{(B=20)}(r)$

Q15. GATE CSE 1994

Open solution

An instance of a relational scheme R(A, B, C) has distinct values for attribute A. Can you conclude that A is a candidate key for R?

(A) TRUE

(B) FALSE

Q16. GATE CSE 1994

Open solution

Give a relational algebra expression using only the minimum number of operators from $(\cup, -)$ which is equivalent to $R \cap S$.

(A) TRUE

(B) FALSE

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Marks 2 23 questions

Q17. GATE CSE 2026 Set 1

Consider a relational database schema with two relations $R(P, Q)$ and $S(X, Y)$.

Let $E = \{\langle u \rangle \mid \exists v \exists w \langle u, v \rangle \in R \wedge \langle v, w \rangle \in S\}$ be a tuple relational calculus expression. Which one of the following relational algebraic expressions is equivalent to E ?

Open solution

(A) $\Pi_P\left(R \bowtie_{R.P=S.X} S\right)$

(B) $\Pi_P\left(S \bowtie_{S.X=R.Q} R\right)$

(C) $\Pi_P\left(R \bowtie_{R.P=S.Y} S\right)$

(D) $\Pi_P\left(S \bowtie_{S.Y=R.Q} R\right)$

Q18. GATE CSE 2025 Set 1

Consider two relations describing teams and players in a sports league:

- teams(tid, tname): tid, tname are team-id and team-name, respectively.
- players(pid, pname, tid): pid, pname, and tid denote player-id, player-name and the team-id of the player, respectively.

Open solution

Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having tname as ' MI '?

(A) $\{$ p.pname $\mid p \in$ players $\wedge \exists t(t \in$ teams $\wedge$ p.tid $=$ t.tid $\wedge$ t.name $=' \mathrm{MI}')\}$

(B) $\{$ p.pname $\mid p \in$ teams $\wedge \exists t(t \in$ players $\wedge$ p.tid $=$ t.tid $\wedge$ t.name $=' \mathrm{Ml}') \backslash\}$

(C) $\{$ p.pname $\mid p \in$ players $\wedge \exists t(t \in$ teams $\wedge$ t.name $=' M1')\}$

(D) $\{p.pname \mid p \in$ teams $\wedge \exists t(t \in$ players $\wedge$ t.name $=$ ' Ml ' $)\}$

Q19. GATE CSE 2024 Set 2

The relation schema, **Person**(pid, $city$), describes the city of residence for every person uniquely identified by pid. The following relational algebra operators are available: *selection, projection, cross product, and rename*.

Open solution

To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of *cross product* operations that must be used is

(A) 1 (B) 2 (C) 3 (D) 4

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Q20. GATE CSE 2019

Open solution

Consider the following relations P(X,Y,Z), Q(X,Y,T) and R(Y,V).

P		
X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Q		
X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

R	
Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

How many tuples will be returned by the following relational algebra query?

$$\prod_X (\sigma_{(P.Y=R.Y \wedge R.V=V2)}(P \times R)) - \prod_X (\sigma_{(Q.Y=R.Y \wedge Q.T>2)}(Q \times R))$$

Q21. GATE CSE 2018

Open solution

Consider the relations $r(A, B)$ and $s(B, C)$, where $s.B$ is a primary key and $r.B$ is a foreign key referencing $s.B$. Consider the query

$Q: \quad r \bowtie (\sigma_{B<5}(s))$

Let LOJ denote the natural left outer-join operation. Assume that r and s contain no null values.

Which one of the following queries is NOT equivalent to Q?

(A) $\sigma_{B<5}(r \bowtie s)$

(B) $\sigma_{B<5}(r \ LOJ \ s)$

(C) $r \ LOJ \ (\sigma_{B<5}(s))$

(D) $\sigma_{B<5}(r) LOJ \ s$

Q22. GATE CSE 2015 Set 2

Open solution

Consider two relations $R_1(A, B)$ with the tuples $(1, 5), (3, 7)$ and $R_2(A, C) = (1, 7), (4, 9)$. Assume that $R(A, B, C)$ is the full natural outer join of R_1 and R_2. Consider the following tuples of the form (A, B, C) : $a = (1, 5, null), b = (1, null, 7), c = (3, null, 9), d = (4, 7, null), e = (1, 5, 7), f = (3, 7, null), g = (4, null, 9)$. Which one of the following statements is correct?

(A) R contains a, b, e, f, g but not c, d.

(B) R contains all of a, b, c, d, e, f, g.

(C) R contains e, f, g but not a, b.

(D) R contains e but not f, g.

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Q23. GATE CSE 2014 Set 2

Open solution

Consider a join (relation algebra) between relations **r(R)**and **s(S)** using the nested loop method. There are 3 buffers each of size equal to disk block size, out of which one buffer is reserved for intermediate results. Assuming **size(r(R)) < size(s(S))**, the join will have fewer number of disk block accesses if

(A) relation r(R) is in the outer loop.

(B) relation s(S) is in the outer loop.

(C) join selection factor between r(R) and s(S) is more than 0.5.

(D) join selection factor between r(R) and s(S) is less than 0.5.

Q24. GATE CSE 2014 Set 3

Open solution

Consider the relational schema given below, where **eId** of the relation **dependent** is a foreign key referring to **empId** of the relation **employee**. Assume that every employee has at least one associated dependent in the dependent relation:

employee (empId, empName, empAge)

dependent (depId, eId, depName, depAge)

Consider the following relational algebra query:

$\Pi_{empId}\ (employee) - \Pi_{empId}\ (employee \bowtie_{(empId=eID)\wedge(empAge\leq depAge)}\ dependent)$

The above query evaluates to the set of empIds of employees whose age is greater than that of

(A) some dependent.

(B) all dependents.

(C) some of his/her dependents.

(D) all of his/her dependents.

Q25. GATE CSE 2012

Open solution

Suppose R_1 (A, B) and R_2 (C, D) are two relation schemas. Let r_1 and r_2 be the corresponding relation instances. B is a foreign key that refers to C in R_2. If data in r_1 and r_2 satisfy referential integrity constraints, which of the following is **ALWAYS TRUE**?

(A) $\prod_B(r_1) - \prod_C(r_2) = \varnothing$

(B) $\prod_C(r_2) - \prod_B(r_1) = \varnothing$

(C) $\prod_B(r_1) = \prod_C(r_2)$

(D) $\prod_B(r_1) - \prod_C(r_2) \neq \varnothing$

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Q26. GATE CSE 2012

Open solution

Consider the following relations A, B and C:

A

ID	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B

ID	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C

ID	Phone	Area
10	2200	02
99	2100	01

How many tuples does the result of the following relational algebra expression contain? Assume that the schema of A ∪ B is the same as that of A. $(A \cup B) \bowtie_{A.Id>40 \vee C.Id<15} C$

(A) 7 (B) 4 (C) 5 (D) 9

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Q27. GATE CSE 2012

Open solution

Consider the following relations A, B and C:

A

ID	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B

ID	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C

ID	Phone	Area
10	2200	02
99	2100	01

How many tuples does the result of the following SQL query contain?

```
SELECT A.Id
FROM A
WHERE A.Age > ALL (SELECT B.Age
                   FROM B
                   WHERE B.Name = 'Arun')
```

(A) 4 (B) 3 (C) 0 (D) 1

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Q28. GATE CSE 2011

Open solution

Consider a relational table r with sufficient number of records, having attributes A_1, A_2,....., A_n and let $1 \leq p \leq n$. Two queries Q1 and Q2 are given below.

$Q1 : \pi_{A_1,\ldots,A_p}\left(\sigma_{A_p=c}\left(r\right)\right)$ where is a constant

$Q2 : \pi_{A_1,\ldots,A_p}\left(\sigma_{c_1 \leq A_p \leq c_2}\left(r\right)\right)$ where c_1 and c_2 are constants

The database can be configured to do ordered indexing on A_p or hashing on A_p. Which of the following statements is TRUE?

(A) Ordered indexing will always outperform hashing for both queries

(B) Hashing will always outperform ordered indexing for both queries

(C) Hashing will outperform ordered indexing on Q1, but not on Q2

(D) Hashing will outperform ordered indexing on Q2, but not on Q1

Q29. GATE CSE 2010

Open solution

The following functional dependencies hold for relations R(A, B, C) and S(B, D, E):

$$B \rightarrow A$$
$$A \rightarrow C$$

The relation R contains 200 tuples and the relation S contains 100 tuples. What is the maximum number of tuples possible in the natural join R $\bowtie$ S?

(A) 100 **(B)** 200 **(C)** 300 **(D)** 2000

Q30. GATE CSE 2008

Open solution

Let R and S be two relations with the following schema

R (P, Q, R1, R2, R3)

S (P, Q, S1, S2)

Where {P, Q} is the key for both schemas. Which of the following queries are equivalent?

I. $\Pi_P\left(R \bowtie S\right)$

II. $\Pi_P\left(R\right) \bowtie \Pi_P\left(S\right)$

III. $\Pi_P\left(\Pi_{P,Q}\left(R\right) \cap \Pi_{P,Q}\left(S\right)\right)$

IV. $\Pi_P\left(\Pi_{P,Q}\left(R\right) - \left(\Pi_{P,Q}\left(R\right) - \Pi_{P,Q}\left(S\right)\right)\right)$

(A) Only I and II **(B)** Only I and III

(C) Only I, II and III **(D)** Only I, III and IV

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Q31. GATE CSE 2007

Open solution

Information about a collection of students is given by the relation **studInfo(studId, name, sex)**. The relation **enroll(studId, courseId)** gives which student has enrolled for (or taken) what course(s). Assume that every course is taken by at least one male and at least one female student. What does the following relational algebra expression represent?

$$\prod_{courseId}((\prod_{studId}(\sigma_{sex='female'}(studInfo)) \times \prod_{courseId}(enroll)) - enroll)$$

(A) Courses in which all the female students are enrolled.

(B) Courses in which a proper subset of female students are enrolled.

(C) Courses in which only male students are enrolled.

(D) None of the above.

Q32. GATE CSE 2007

Open solution

Consider a selection of the form $\sigma_A \leq 100(r)$, where r is a relation with 1000 tuples. Assume that the attribute values for A among the tuples are uniformly distributed in the interval [0, 500]. Which one of the following options is the best estimate of the number of tuples returned by the given selection query?

(A) 50 (B) 100 (C) 150 (D) 200

Q33. GATE CSE 2007

Open solution

Consider the relation **employee(name, sex, supervisorName)** with name as the key, supervisorName gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

$$\{e.name \mid employee(e) \wedge (\forall x)[\neg employee(x) \vee x.supervisorName \neq e.name \vee x.sex =' male']\}$$

(A) Names of employees with a male supervisor.

(B) Names of employees with no immediate male subordinates.

(C) Names of employees with no immediate female subordinates.

(D) Names of employees with a female supervisor.

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Q34. GATE CSE 2007

Open solution

Consider the following relation schemas :
b-Schema = (b-name, b-city, assets)
a-Schema = (a-num, b-name, bal)
d-Schema = (c-name, a-number)

Let branch, account and depositor be respectively instances of the above schemas. Assume that account and depositor relations are much bigger than the branch relation.

Consider the following query:
$\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"} \wedge \text{bal} < 0}$ (branch $\bowtie$ (account $\bowtie$ depositor))
Which one of the following queries is the most efficient version of the above query ?

(A) $\Pi_{\text{c-name}}$ ($\sigma_{\text{bal} < 0}$ ($\sigma_{\text{b-city = "Agra"}}$ branch $\bowtie$ account) $\bowtie$ depositor)

(B) $\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"}}$branch $\bowtie$ ($\sigma_{\text{bal} < 0}$ account $\bowtie$ depositor))

(C) $\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"}}$ branch $\bowtie$ $\sigma_{\text{b-city = "Agra"} \wedge \text{bal} < 0}$ account) $\bowtie$ depositor)

(D) $\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"} \wedge \text{bal} < 0}$ account $\bowtie$ depositor))

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Q35. GATE CSE 2005

Open solution

A table **'student'** with schema (roll, name, hostel, marks), and another table **'hobby'** with schema (roll, hobbyname) contains records as shown below:

Table: student

Roll	Name	Hostel	Marks
1798	Manoj Rathod	7	95
2154	Soumic Banerjee	5	68
2369	Gumma Reddy	7	86
2581	Pradeep Pendse	6	92
2643	Suhas Kulkarni	5	78
2711	Nitin Kadam	8	72
2872	Kiran Vora	5	92
2926	Manoj Kunkalikar	5	94
2959	Hemant Karkhanis	7	88
3125	Rajesh Doshi	5	82

Table: hobby

Roll	Hobbyname
1798	chess
1798	music
2154	music
2369	swimming
2581	cricket
2643	chess
2643	hockey
2711	volleyball
2872	football
2926	cricket
2959	photography
3125	music
3125	chess

The following SQL query is executed on the above tables:

```
Select hostel
From student natural join hobby
Where marks >= 75 and roll between 2000 and 3000;
```

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Relations S and H with the same schema as those of these two tables respectively contain the same information as tuples. A new relation S' is obtained by the following relational algebra operation:

$S' = \prod_{hostel}((\sigma_{S.roll=H.roll}$
$(\sigma_{marks>75 \ \ and \ \ roll>2000 \ \ and \ \ roll<3000}(S))X(H))$

The difference between the number of rows output by the SQL statement and the number of tuples in S' is

(A) 6 (B) 4 (C) 2 (D) 0

Q36. GATE CSE 2004

Consider the relation **Student (name, sex, marks)**, where the primary key is shown underlined, pertaining to students in a class that has at least one boy and one girl. What does the following relational algebra expression produce? (Note: ρ is the rename operator).

$$\prod_{name}\left(\sigma_{sex=female}\left(Student\right)\right) - \prod_{name}\left(Student \bowtie \rho_{n,x,m}\left(Student\right)\right)$$
$$\begin{pmatrix} sex = female \\ \wedge x = male \\ \wedge marks \leq m \end{pmatrix}$$

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Open solution

(A) names of girl students with the highest marks
(B) names of girl students with more marks than some boy student
(C) names of girl students with marks not less than some boy student
(D) names of girl students with more marks than all the boy students

Q37. GATE CSE 2001

Which of the following relational calculus expressions is not safe?

(A) $\{t \mid \exists u \in R_1\,(t[A] = u[A]) \wedge \neg\exists s \in R_2\,(t[A] = s[A])\}$
(B) $\{t \mid \forall u \in R_1\,(u[A] = x \Rightarrow \exists s \in R_2\,(t[A] = s[A] \wedge s[A] = u[A]))\}$
(C) $\{t \mid \neg(t \in R_1)\}$
(D) $\{t \mid \exists u \in R_1\,(t[A] = u[A]) \wedge \exists s \in R_2\,(t[A] = s[A])\}$

Open solution

Q38. GATE CSE 1998

Which of the following query transformations (i.e. replacing the l.h.s. expression by the r.h.s. expression) is incorrect? R_1 and R_2 are relations, C_1, C_2 are selection conditions and A_1, A_2 are attributes of R_1?

(A) $\sigma_{C_1}(\sigma_{C_2}(R_1)) \rightarrow \sigma_{C_2}(\sigma_{C_1}(R_1))$
(B) $\sigma_{C_1}(\pi_{A_1}(R_1)) \rightarrow \pi_{A_1}(\sigma_{C_1}(R_1))$
(C) $\sigma_{C_1}(R_1 \cup R_2) \rightarrow \sigma_{C_1}(R_1) \cup \sigma_{C_1}(R_2)$
(D) $\pi_{A_1}(\sigma_{C_1}(R_1)) \rightarrow \sigma_{C_1}(\pi_{A_1}(R_1))$

Open solution

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Q39. GATE CSE 1996

Open solution

A library relational database system uses the following schema

USERS (User #, User Name, Home Town)

BOOKS (Books # Book Title, Author Name)

ISSUED (Book #, User #, Date)

Explain in one English sentence, what each of the following relational algebra queries is designed to determine.

(a) $\sigma_{\text{User\#}=6}\left(\pi_{\text{User\#, Book Title}}\left((\text{USERS} \bowtie \text{ISSUED}) \bowtie \text{BOOKS}\right)\right)$

(b) $\pi_{\text{Author Name}}\left(\text{BOOKS} \bowtie \sigma_{\text{Home Town=Delhi}}\left(\text{USERS} \bowtie \text{ISSUED}\right)\right)$

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Answer Key

Marks 1

1. **2**	2. **A, B**	3. **819 to 820**	4. **A**	5. **C**
6. **A**	7. **D**	8. **C**	9. **B**	10. **A**
11. **C**	12. **C**	13. **B**	14. **C**	15. **B**
16. **B**				

Marks 2

17. **B**	18. **A**	19. **B**	20. **1**	21. **C**
22. **C**	23. **A**	24. **D**	25. **A**	26. **A**
27. **B**	28. **C**	29. **A**	30. **D**	31. **B**
32. **D**	33. **C**	34. **A**	35. **B**	36. **D**
37. **C**	38. **D**	39. **(a) Select the titles of the books issued to User# = 6 (b) Select author names of the books issued to users who are from Delhi.**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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